La Scuola d'Italia / Dr. Huson / IB Math 11
28 October 2026 Markscheme

2.6 Early Finishers: Lines, Exponents, Radicals and Sequences — Markscheme

Extension set, scored separately from the quiz. Calculator allowed. Version (a) repeats problems 3 and 4 of quiz version (b), with the same marking.

Version (a)

Total: 19 marks.

1. [Maximum mark: 11]

(a) $(5^2)^{x+1} = (5^3)^{2x-3}$ **(M1)**

$2x + 2 = 6x - 9$ **A1**

$x = \dfrac{11}{4}$ **A1**

Accept 2.75. *M1 for writing both sides as powers of* 5*; it may be implied by a correct equation in the exponents. Accept* $2(x+1) = 3(2x-3)$ *for the first A1. A bare* $x = \frac{11}{4}$ *earns M0A0A1. Wrong methods:* $2x + 1 = 6x - 9$ *or* $2x + 2 = 6x - 3$ *(a bracket not multiplied out) gives* $\frac{5}{2}$ *or* $\frac{5}{4}$*: M1A0A0.*

(b) $2a + 2b\sqrt{3} + a\sqrt{3} + 3b = 12 + 7\sqrt{3}$ **M1**

$2a + 3b = 12, \quad a + 2b = 7$ **A1**

$a = 3, \quad b = 2$ **A1**

Accept the conjugate route: $\frac{12+7\sqrt{3}}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{24-12\sqrt{3}+14\sqrt{3}-21}{1} = 3 + 2\sqrt{3}$ *(M1A1A1). Accept trial with the expansion of* $(2+\sqrt{3})(3+2\sqrt{3})$ *shown as a check (M1A1A1). A bare answer earns M0A0A1.*

(c) $9^x = (3^x)^2 = u^2$, so $3u^2 - 82u + 27 = 0$ **M1**

$(3u - 1)(u - 27) = 0$ **(M1)**

$u = \dfrac{1}{3}$ or $u = 27$ **A1**

$3^x = \dfrac{1}{3} \Rightarrow x = -1$ **A1**

$3^x = 27 \Rightarrow x = 3$ **A1**

M1 for the quadratic in u*. (M1) for an attempt at any method of solving their quadratic (factorizing, the formula, completing the square or the GDC); it may be implied by a correct value of* u*. A1 for either* $u = \frac{1}{3}$ *or* $u = 27$ *correct. Stopping at* $u = \frac{1}{3}, 27$ *earns M1(M1)A1A0A0. Wrong method: reading* 3^x *as* $3x$ *gives* $x = \frac{1}{9}, 9$*: A0A0 for those values.*

2. [Maximum mark: 8]

(a) $u_n = 7 + 3(n-1)$ **A1**

$v_n = 2(1.5)^{n-1}$ **A1**

Accept $u_n = 3n + 4$.

(b) $2(1.5)^{n-1} > 7 + 3(n-1)$ **(M1)**

$u_7 = 25$, $v_7 = 22.8$; $\quad u_8 = 28$, $v_8 = 34.2$ **A1**

$n = 8$ **A1**

M1 for the inequality, or a GDC table or graph of both sequences; it may be implied by the values on both sides of the change. A1 for the values on both sides of the change. A bare $n = 8$ *earns M0A0A1. FT from (a):* $v_n = 2(1.5)^n$ *gives* $n = 7$.

(c) $\dfrac{2\left(1.5^n - 1\right)}{1.5 - 1} > \dfrac{n}{2}\left(2(7) + 3(n-1)\right)$ **M1**

$n = 9$: $149.8 < 171$; $\quad n = 10$: $226.7 > 205$ **A1**

$n = 10$ **A1**

M1 for both sum formulas (or a GDC table of both sums). A1 for the sums on both sides of the change. Wrong method: comparing terms instead of sums gives $n = 8$*: M0A0A0. FT from (a).*

Version (b)

Total: 22 marks.

1. [Maximum mark: 10]

(a) gradients $-\frac{k}{2}$ and $-\frac{3}{k-1}$ **A1**

$-\frac{k}{2} = -\frac{3}{k-1}$ **M1**

$k(k-1) = 6, \quad k^2 - k - 6 = 0, \quad (k-3)(k+2) = 0$

$k = 3$ **A1**

$k = -2$ **A1**

Accept comparing coefficients, $\frac{k}{3} = \frac{2}{k-1}$, for the A1M1. Finding $k = 3$ by inspection only, with no equation, earns A0M0A1A0.

(b) $\left(-\frac{k}{2}\right)\left(-\frac{3}{k-1}\right) = -1$ **M1**

$3k = -2(k-1)$ **A1**

$k = \frac{2}{5}$ **A1**

Accept 0.4. *Wrong method: product of the gradients $= +1$ gives $k = -2$: M0A0A0.*

(c) L_1: $x + 2y = 7, \quad L_2$: $3x = 5$ **M1**

$x = \frac{5}{3}$ **A1**

$y = \frac{8}{3}$ **A1**

M1 for substituting $k = 1$ into both equations (L_2 is vertical). Accept 1.67 *and* 2.67.

2. [Maximum mark: 12]

(a) $\dfrac{k+1}{k-3} = \dfrac{3k-1}{k+1}$ **M1**

$k^2 + 2k + 1 = 3k^2 - 10k + 3$ **A1**

$2k^2 - 12k + 2 = 0, \quad k^2 - 6k + 1 = 0$ **AG**

M1 for equating the two ratios, or $(k+1)^2 = (k-3)(3k-1)$*. A1 for both sides expanded correctly.*

(b) $k = \dfrac{6 \pm \sqrt{36-4}}{2} = \dfrac{6 \pm 4\sqrt{2}}{2}$ **M1**

$k = 3 + 2\sqrt{2}$ or $k = 3 - 2\sqrt{2}$ **A1**

A1 for both values in exact form. 5.83 *and* 0.172 *earn M1A0.*

(c) $r = \dfrac{k+1}{k-3} = \dfrac{4+2\sqrt{2}}{2\sqrt{2}}$ **M1A1**

$= \dfrac{\left(4+2\sqrt{2}\right)\sqrt{2}}{4} = \dfrac{4\sqrt{2}+4}{4}$ **M1**

$r = 1 + \sqrt{2}$ **A1**

First M1 for the ratio of consecutive terms, either pair. Second M1 for rationalizing. Wrong method: the inverted ratio $\frac{k-3}{k+1}$ *gives* $\sqrt{2}-1$*: M0A0M1A0. FT from (b).*

(d) $r = \dfrac{4-2\sqrt{2}}{-2\sqrt{2}} = 1 - \sqrt{2}$ **A1**

$|r| = 0.414\ldots < 1$, so the sum to infinity exists **R1**

$S_\infty = \dfrac{-2\sqrt{2}}{1-\left(1-\sqrt{2}\right)} = \dfrac{-2\sqrt{2}}{\sqrt{2}}$ **M1**

$S_\infty = -2$ **A1**

R1 for $|r| < 1$ *(or* $-1 < r < 1$*) with their value of* r*. Wrong method: the inverted ratio gives* $r = -1 - \sqrt{2}$ *(no sum to infinity) or uses* $\sqrt{2} - 1$ *to get* $2 + 2\sqrt{2}$*: A0, R1 only if their* $|r| < 1$*, M1A0. FT from (b).*